CLASS AVES

Birds are an amazing and diverse group of animals. They are best known for their ability to fly.

Birds are a diverse group of vertebrates that evolved from reptiles during the Mesazoic Era.

Today estimated 300 billion birds belonging to more than 9700 species inhabit on the planet.

Origin and Early Evolution

- Evidence from fossils and from studies of comparative anatomy indicates that birds evolved from reptiles

- The fossil genus Archaeopteryx link between reptiles and birds.

- Live 145 million years ago

The evolution of birds began in the Jurassic Period, with the earliest

birds derived from a clade of theropoda dinosaurs named Paraves

Caudipteryx

The “fuzzy raptor”

The discovery of feathered dinosaurs in Liaoning, China, has excited the many paleontologists who suspected a direct link between dinosaurs and birds.

Archaeopteryx- First significant specimen

The first fossil of *Archaeopteryx* was a single feather, found in 1860.

This feather was not only exceptionally preserved, but showed the asymmetric form that is characteristic of flight feathers.

Archaeopteryx- the London specimen

This specimen found in 1861, called "The London Specimen" was significant in that it established the type of bird from which the single feather found the previous year was derived.

(Note there are so few specimens of *Archaeopteryx* that they are named for the museums/cities in which they are curated).

The London Specimen resides in the British Museum in London, U.K.

Berlin Specimen

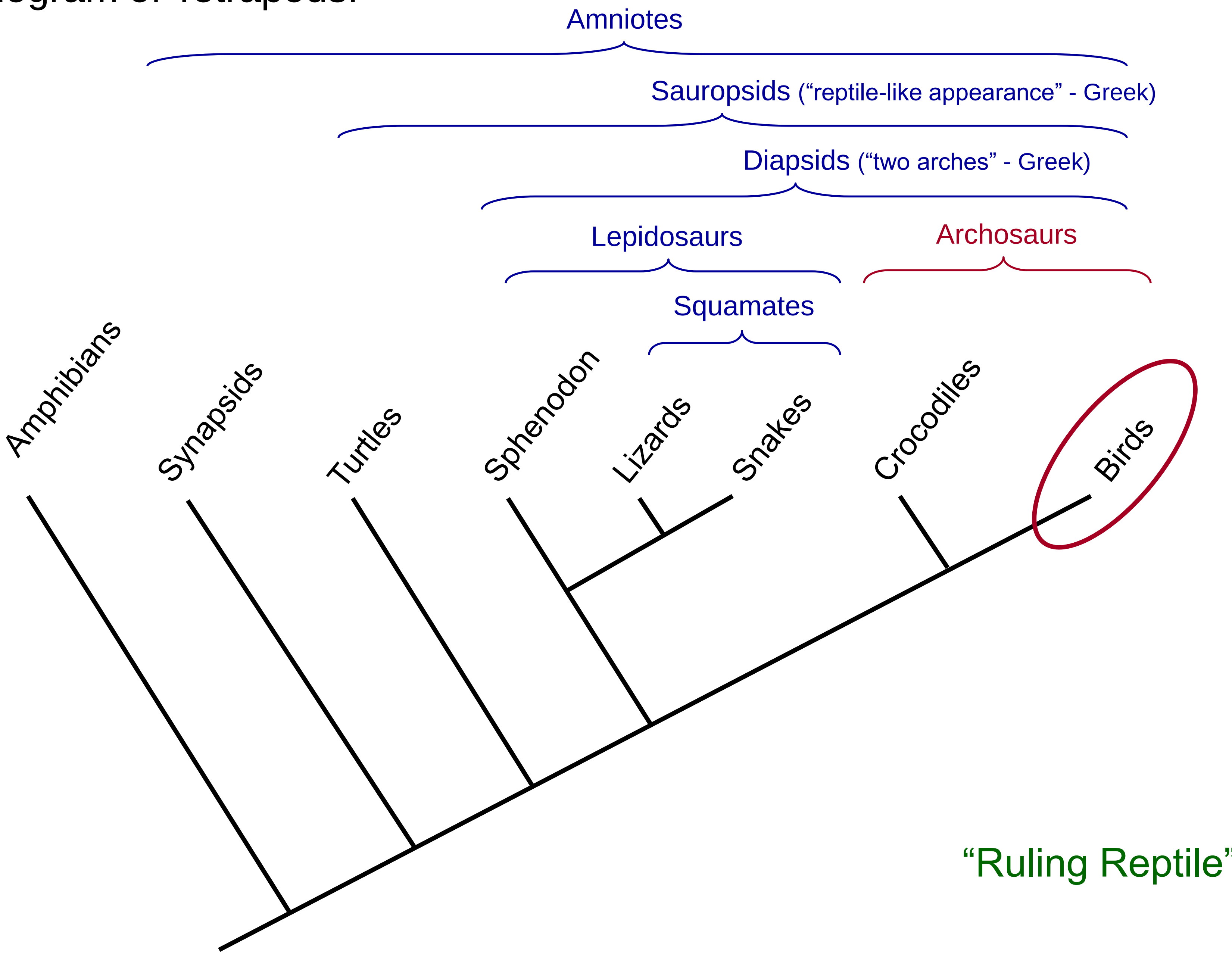
By far the most famous and best preserved specimen of Archaeopteryx is the Berlin specimen discovered in 1877 (curated in the Humboldt University Museum of Natural History in Berlin)

This specimen shows most of the significant features that are considered evidence of a dinosaur-bird connection.

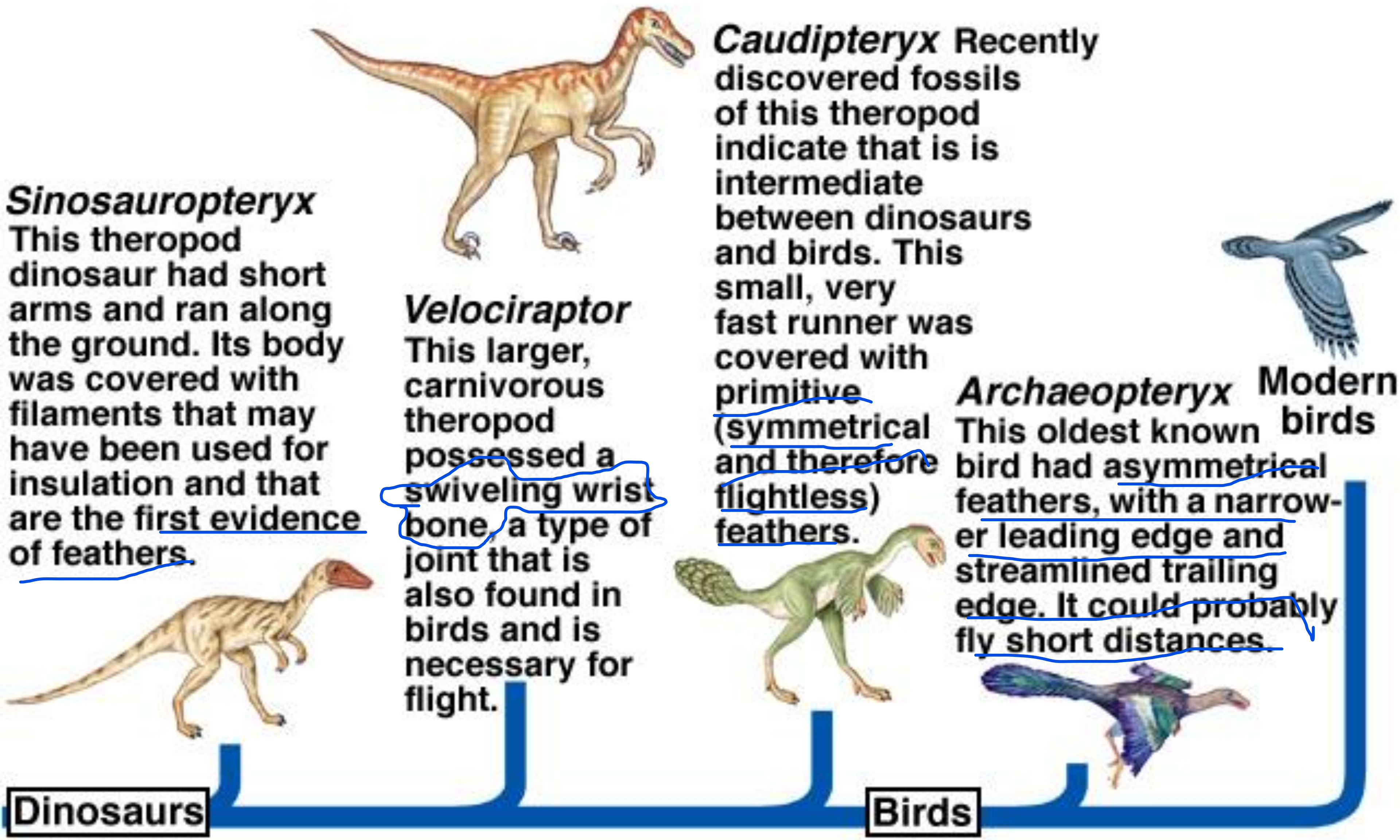
Archaeopteryx lithographica

Reptiles

Cladogram of Tetrapods:



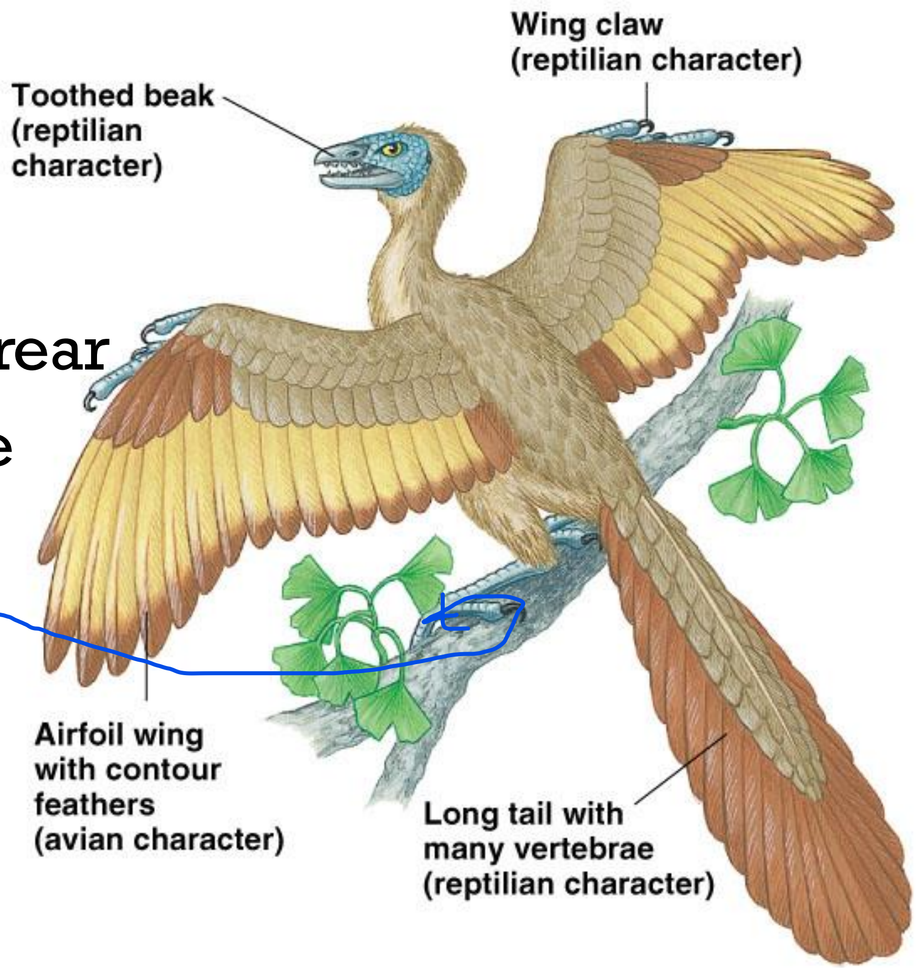
Evolutionary Path to the Birds



Archeopteryx

Reptilian Features:

1) Non-keratinized snout
    - Teeth present
2) Trunk vertebrae not fused
3) Neck attaches to skull from rear
4) Long tail with free vertebrae
5) Ankle / wrist bones free



Avian Features:

1) Feathers
2) Opposable big toe (hallux)
3) Furcula (wishbone) present

http://huehueteotl.files.wordpress.com/2007/06/archaeopteryx.jpg

Hesperornis is a genus of flightless aquatic birds that spanned the first half of the Campanian age of the Late Cretaceous period.

Characteristics of Class Aves

Key Points

- Head, trunk, tail
- Wings, 2 legs
- Feathers
- Bones – lightweight
- Teeth – none
- Air sacs
- Endothermic
- Heart – four chambers
- Amniotic egg

Characteristics of Class Aves *continue*

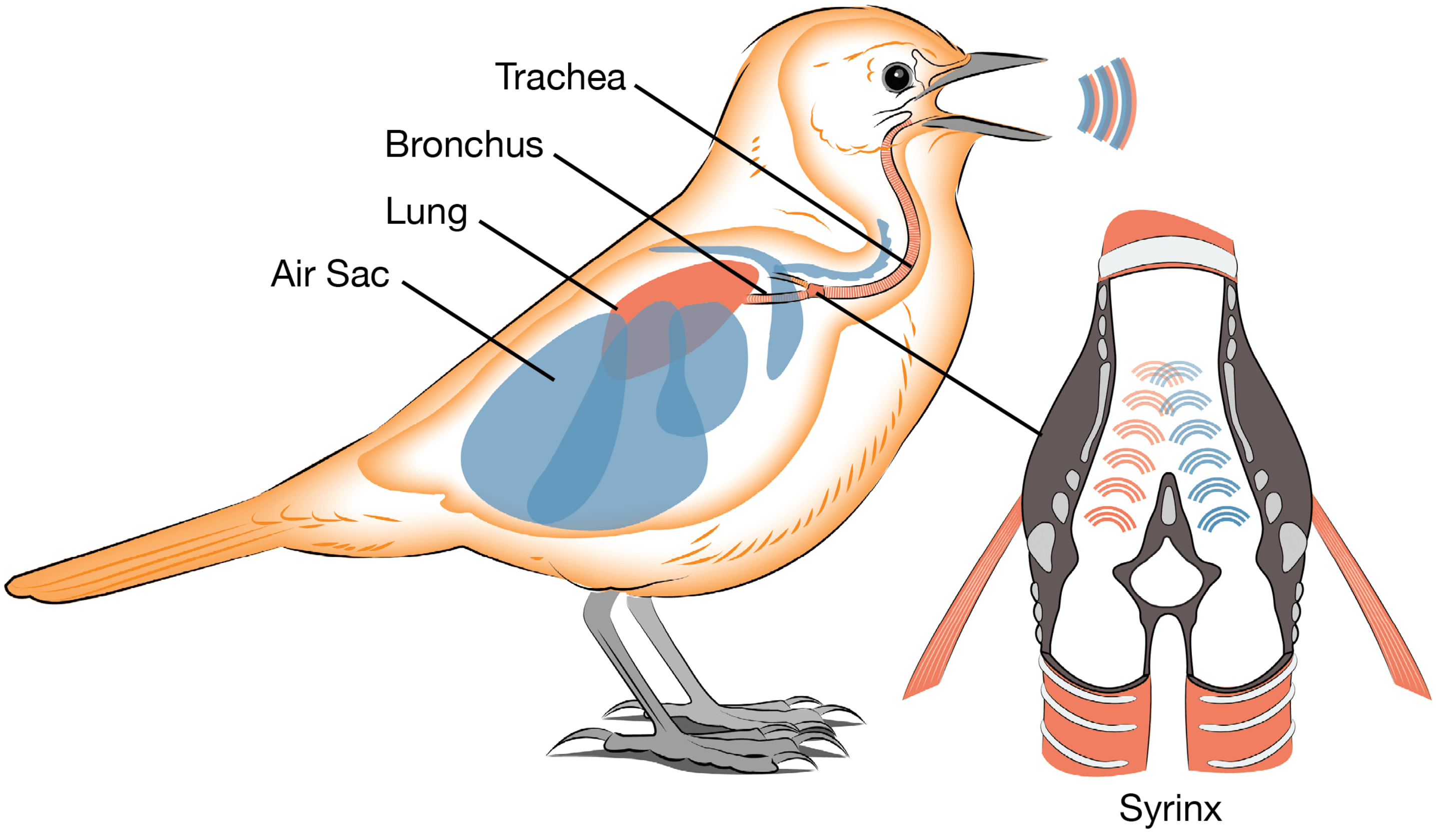
- Body usually spindle with four divisions; Head, neck, trunk and tail.
- Neck is disproportionately long for balancing and food gathering.
- Limbs paired with the forelimbs usually modified for flying. Posterior pair adapted for Perching, walking or swimming.
- Epidermal converting of **feathers** and leg **scales**. No sweat glands, oil or preen gland present in the tail.

Characteristics of Class Aves *continue*

- Fully ossified skeleton with air cavities. Skulled bones fused with one **occipital condyle**. No **teeth**, each jaw covered with a horny sheath forming a beak. **Sternum** is well develop with or without a keel. Single bone in **middle ear**

- Endothermic

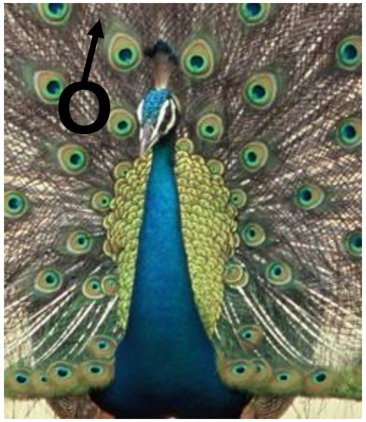
- Respiration by slightly expansible lungs with thin **air sacs** among the visceral and skeleton. **Syrinx (voice box)** near junction of trachea and bronchi.


Trachea
Bronchus
Lung
Air Sac
Syrinx

Characteristics of Class Aves *continue*

- Excretory system of metanephric kidney, ureters open into cloaca, **no bladder**. Main nitrogenous waste is uric acid.

- Sexes are separate

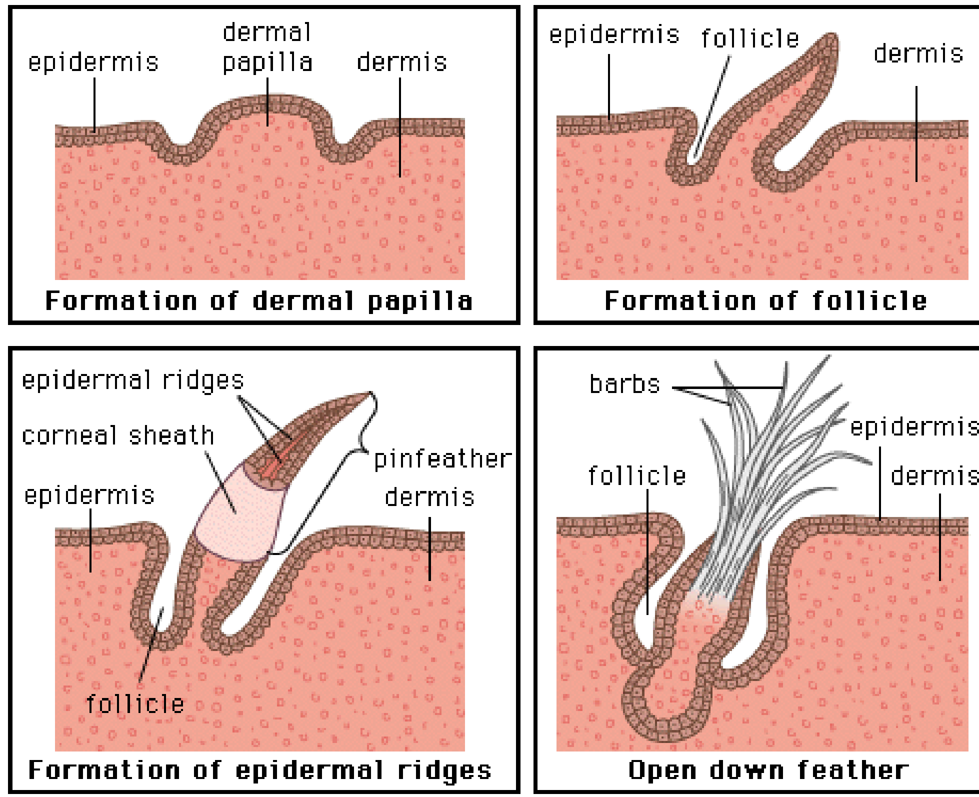


- Fertilizations internal; amniotic eggs with much yolk and hard calcareous shells; embryonic membranes in eggs during development; incubation external.

Feathers

Feathers develop in a fashion similar to the epidermal scales of reptiles, and this similarity is one source of evidence that demonstrates the evolutionary ties between reptiles and birds.

Only the inner pulp of feathers contains dermal elements, such as blood vessels which supply nutrients and pigments for the growing feather. As feathers mature their blood supply is cut off and the feather becomes dead, keratinized.



Feathers:

- Similar to reptilian scales (beta-keratin – present in birds / reptiles)
  - Retain scales on non-feathered parts

- Dead structures; damage repair = replacement
  - Specialized pockets of epidermal / dermal cells (follicles)

➢ Feathers appear in fossil record long before flight (*e.g.*, *Caudipteryx*)

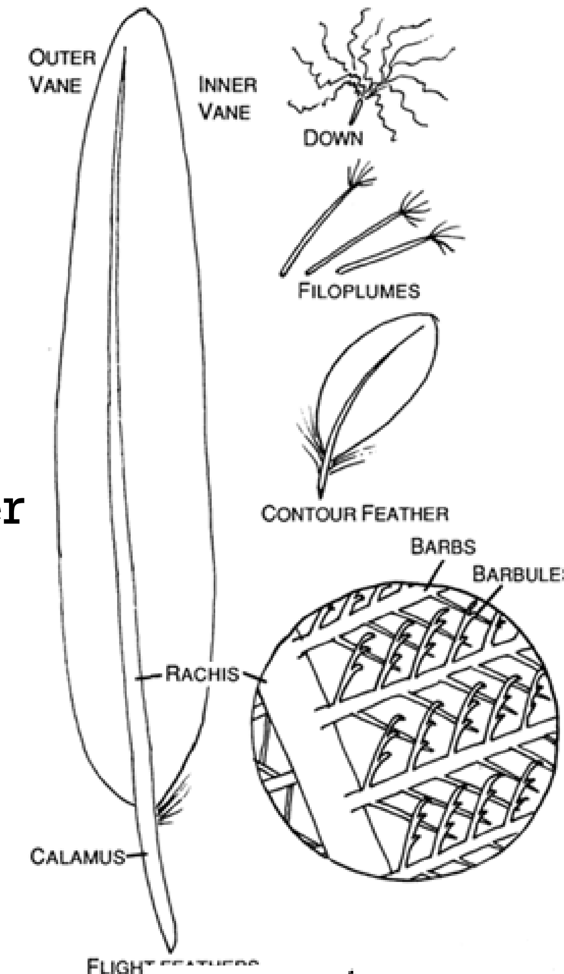
- Hypotheses: 1) Insulation to retain heat (not endothermic…)
  2) Social interactions (e.g., reproduction)
  3) Shading / insulation for eggs

- Current Functions of Feathers:

Feathers:

A) Contour Feathers (flight feathers – vaned…)

- Long central shaft (rachis)
- Broad vane with numerous barbs
    - Barbules hook barbs together
- Calamus (quill) anchors feather in follicle (skin)
- Mobile – individual muscles control each feather
- Stream-lined; decreases drag
- Asymmetrical & curved (independent airfoils)



B) Filoplumes (Provide sensory information)

- Long rachis with few barbs at end

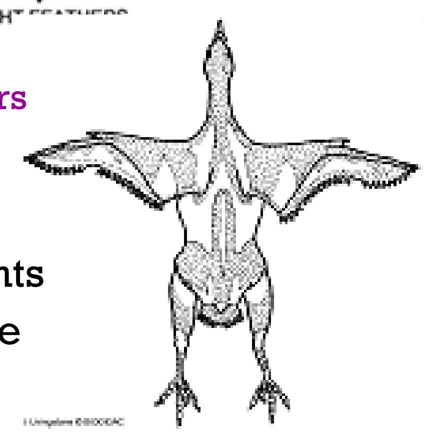
C) Down Feathers (Insulation)

- Lack central shaft; barbs from feather base
- lack barbules; barbs move freely

D) Bristles

2000 – 4000 feathers
(~ 15% BW)

Feather Tracts:
Feather attachments grouped in dense concentrations



- Range from drab to colorful…

A) **Biochrome Pigments**

- Melanins = earth tones (*e.g.*, grays / blacks / browns)
    - Resist feather wear (e.g., black wing tips resist fraying)
    - Resist bacterial degradation (wet climate = dark color)
    - Absorb energy (thermoregulation / feather drying)
- Carotenoids = vibrant colors (*e.g.*, bright yellows / oranges / reds)
    - Derived from diet (honest signaling…)
- Porphyrins = vibrant colors (*e.g.*, bright brown / green / magenta)
    - Chemically related to hemoglobin

B) **Structural Colors**

- Result from physical alteration of light (e.g., iridescences)
    - Nanometer-scale structuring in feather:

Can occur together…

1) Air bubbles = White (unpigmented feathers)

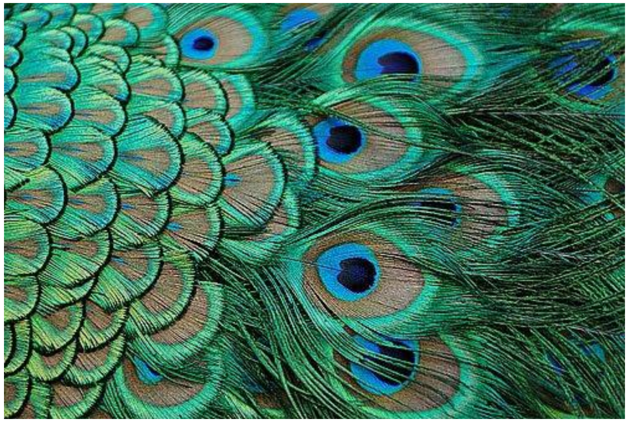
2) Melanin granules (melanosomes) = iridescence

➢ High UV reflectance (birds capable of detecting UV light!)

Feather Colors –

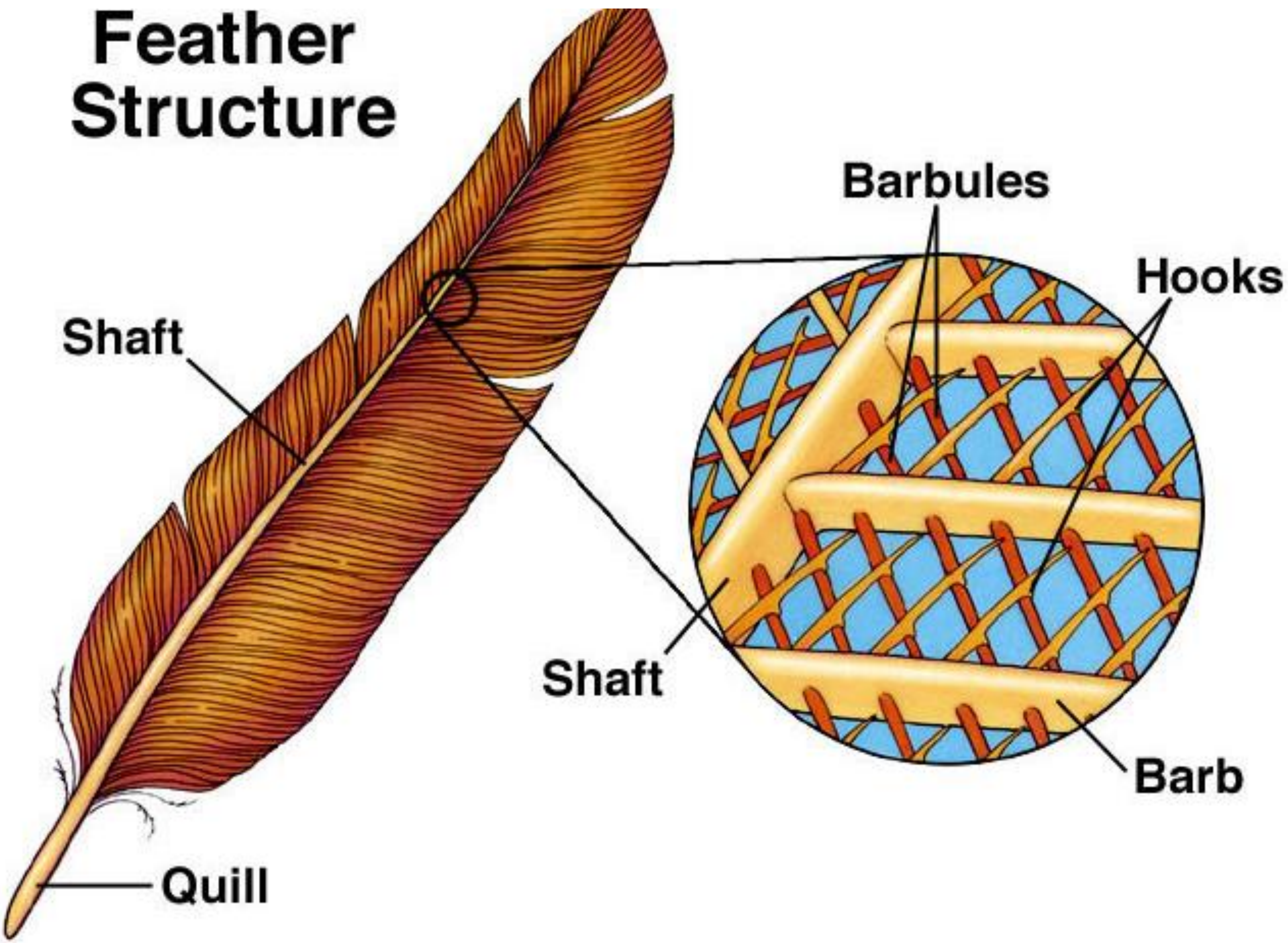
Most of the colors are from pigments deposited in the feather during formation.

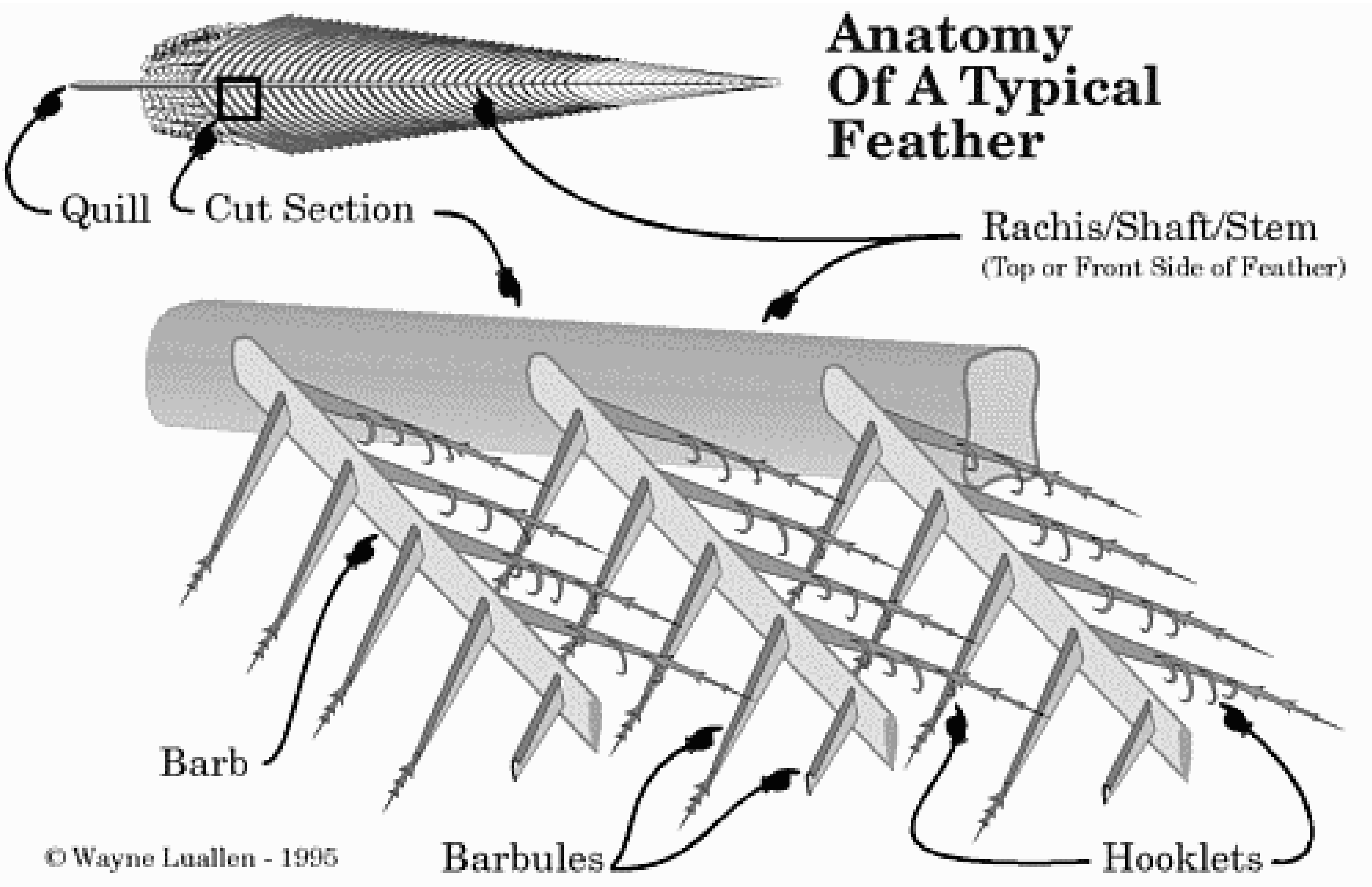
Structural Colors arise from the irregular formation of the surface of the feathers, blue feathers (are never blue from pigment) the irregularities reflect blue light and allow the other colors of white light to pass into the barb and are absorbed by pigments in the barb.



Feather Structure

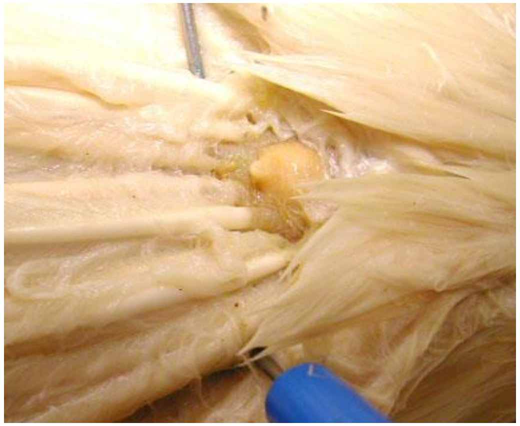
1. **Shaft** - also called RACHIS is the center of a contour feather
2. **Barb** - hair like projection from the shaft arranged parallel to each other
3. **Barbules** - small projections from the barbs the intersect with other barbules
4. **Hooklets** - interlocking projections off barbules
5. **Vane** - consists of hundreds of tiny barbs that branch off of either side of the rachis
6. **Quill** - start from feather shaft


Feather Structure
Barbules
Hooks
Shaft
Shaft
Barb
Quill


Anatomy Of A Typical Feather
Quill
Cut Section
Rachis/Shaft/Stem
(Top or Front Side of Feather)
Barb
Barbules
Hooklets
© Wayne Luallen - 1995

- Feather care important…

  - Uropygial (Preen) Gland:



  - Located at base of rump
  - Secrete rich oil (waxes / fatty acids / fat / water)

  - Preserves feather moistness / flexibility
  - Cleans / waterproofs feathers
  - Regulates bacterial / fungal growth
  - Repel would-be predators (foul-smelling

    - Seasonal display
    - replace feather wear
    - parasite infestations

- Birds go through series of feather coats in lifetime
  - Molt: Replacement of entire plumage (= feather coat)

Pattern of molting varies in taxa

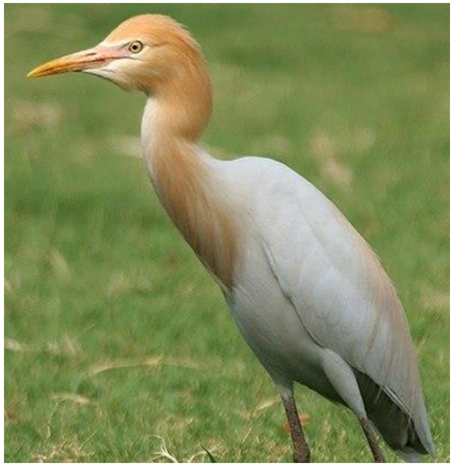
1.bird hatches with downy feathers
2.juvenile feathers grow, molts
3.once sexual maturity is reached a prenuptial molt before breeding season

Seasonal display

Juvenile

Adult

Non Breeding



Breeding